

Reza Baharani

Senior AI Engineer | Health Sensing & On-Device ML | PhD

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🎓 Google Scholar

Experience

ForesightCares Inc., Senior AI Engineer

- Architected and deployed real-time 3D human pose estimation pipeline on iPad achieving 30fps inference for clinical mobility assessments (CDC STEADI: TUG, 30-Second Chair Stand, 4-Stage Balance)
- Engineered on-device AI inference system using CoreML and SceneKit, eliminating cloud dependencies and reducing model bundle size by migrating from third-party ML frameworks to native Apple infrastructure
- Designed multi-stage signal processing pipeline (One Euro filters, Butterworth IIR, median filters) achieving clinical-grade measurement accuracy for joint angle computation and gait analysis
- Built multi-agent AI orchestration system automating regulatory compliance analysis for CMS CY 2026 Medicare RTM billing codes, producing audit-ready documentation, workbooks, and presentations
- Built serverless AI-powered exercise recommendation service on AWS Lambda, leveraging cloud inference and Jinja2-based prompt engineering to deliver personalized mobility programs from a curated library of 41 exercises based on CDC STEADI fall-risk protocols
- Developed Field-of-View calibration model to correct 3D pose estimation across varying camera angles, improving measurement accuracy for at-home and clinical deployment scenarios
- Implemented state machine architecture for 5+ clinical assessment protocols with hysteresis-based transitions, handling edge cases including camera occlusion and multi-person environments
- Co-invented AI-powered mobility assessment system (US Patent 12,343,138 B2), enabling automated fall risk evaluation and real-time exercise coaching via 3D skeleton modeling from monocular video
- Created 3D motion capture dataset using IMU-based mocap system for training and validation, collecting synchronized 60fps video and 400Hz motion data from 100+ subjects to validate pose estimation accuracy

Charlotte, NC
Jan 2022 – present
4 years 8 months

UNC Charlotte, Research Scientist

- Developed self-supervised human motion foundation model using discrete motion tokenization and Vision Transformer pretraining, achieving competitive action classification (89.4%), one-shot recognition (57.4%), and anomaly detection (76.56% AUC-ROC) with a single unified backbone

Charlotte, NC
Oct 2024 – Mar 2025
6 months

UNC Charlotte, Postdoctoral Researcher

- Architected scalable intelligent video surveillance systems for AIoT, including Ancilia and REVAMP2T, enabling privacy-aware multi-camera pedestrian tracking with real-time edge inference across distributed infrastructure (IEEE IoT Journal; 89 and 93 citations)

Charlotte, NC
Aug 2021 – Dec 2022
1 year 5 months

UNC Charlotte, Graduate Research Assistant

Charlotte, NC

Aug 2017 – Aug 2021

4 years 1 month

- Designed algorithm/architecture co-design framework (DeepDive) for deep separable CNNs on Xilinx FPGA, enabling power-efficient edge inference for MobileNetV2 (published at GLSVLSI 2021)
- Developed LSTM-based real-time reliability prediction system for power electronics converters, achieving scalable monitoring on IoT edge devices at 1.87W power budget (published in IEEE IoT Journal, 74 citations)
- Created lightweight temporal convolutional network for vehicle trajectory prediction in highways, matching state-of-the-art accuracy with significantly reduced computational complexity (published in IEEE TITS, 97 citations)
- Co-developed gem5-SALAM system architecture simulator for LLVM-based accelerator modeling (74 citations, IEEE/ACM MICRO 2020)
- Published 12+ peer-reviewed papers across IEEE, ACM venues accumulating 645+ citations (h-index: 11) spanning edge AI, FPGA acceleration, intelligent transportation, and IoT
- Designed attention-based temporal convolutional networks (ATCN) enabling fast time-series classification on resource-constrained embedded devices with formalized hyper-parameters for application-specific architecture tuning (published in ACM TECS)
- Deployed ATCN on Cortex-M7 microcontroller consuming only 49% of 320KB RAM and 15% of 1MB flash; validated across 70 UCR 2018 time-series benchmarks using data augmentation (jittering, magnitude warping, window warping, scaling)
- Developed real-time person re-identification system at the edge using mixed precision (FP16/FP32) inference, achieving efficient tracking on embedded GPUs with MobileNetV2 backbone (published at ICIAR 2019, 17 citations)

Education

PhD University of North Carolina at Charlotte, Electrical and Computer Engineering

Charlotte, NC

Aug 2017 – Aug 2021

- Advisor: Prof. Hamed Tabkhi
- Focus: Computer Architecture and Deep Learning

Publications

MoFM: A Large-Scale Human Motion Foundation Model

Feb 2025

Reza Baharani, G.A. Noghre, A.D. Pazho, G. Maldonado, H. Tabkhi
(arXiv:2502.05432)

DeepTrack: Lightweight Deep Learning for Vehicle Trajectory Prediction in Highways

Jan 2022

V. Katariya, Reza Baharani, et al.
(IEEE Trans. Intelligent Transportation Systems)

REVAMP2T: Real-Time Edge Video Analytics for Multicamera Privacy-Aware Pedestrian Tracking

Jan 2019

C. Neff, M. Mendieta, S. Mohan, Reza Baharani, et al.
(IEEE IoT Journal)

Ancilia: Scalable Intelligent Video Surveillance for the AIoT

Jan 2023

A.D. Pazho, C. Neff, G.A. Noghre, B.R. Ardabili, S. Yao, Reza Baharani, H. Tabkhi
(IEEE IoT Journal)

Real-Time Deep Learning at the Edge for Scalable Reliability Modeling of Si-MOSFET Power Electronics Converters

Jan 2019

Reza Baharani, et al.
(IEEE IoT Journal)

ATCN: Resource-Efficient Processing of Time Series on Edge

Jan 2022

Reza Baharani, H. Tabkhi
(ACM Trans. Embedded Computing Systems (TECS))

Patents

- US 12,343,138 B2: AI Powered Mobility Assessment System, ForesightCares Inc., Jul. 2025. Inventors: H. Tabkhivayghan, M. Azarbayjani, R. Baharani.

Skills

Languages: Python, C/C++, Java, Swift, CUDA

ML/DL Frameworks: PyTorch, TensorFlow, CoreML, ONNX, Vision Transformers

Apple/Mobile: SwiftUI, SceneKit, Combine, CoreML, iOS/iPadOS

DevOps & Cloud: Git, Docker, AWS (Lambda, S3, DynamoDB), Terraform, ROS 2

Edge & Embedded: NVIDIA Jetson, Xilinx FPGA, ARM Cortex-M7, Mixed Precision (FP16/FP32)

Signal Processing: One Euro Filters, Butterworth IIR, Median Filters, Kalman Filtering

Domains: Computer Vision, 3D Pose Estimation, Human Motion Understanding, Self-Supervised Learning, Embedded AI, Signal Processing